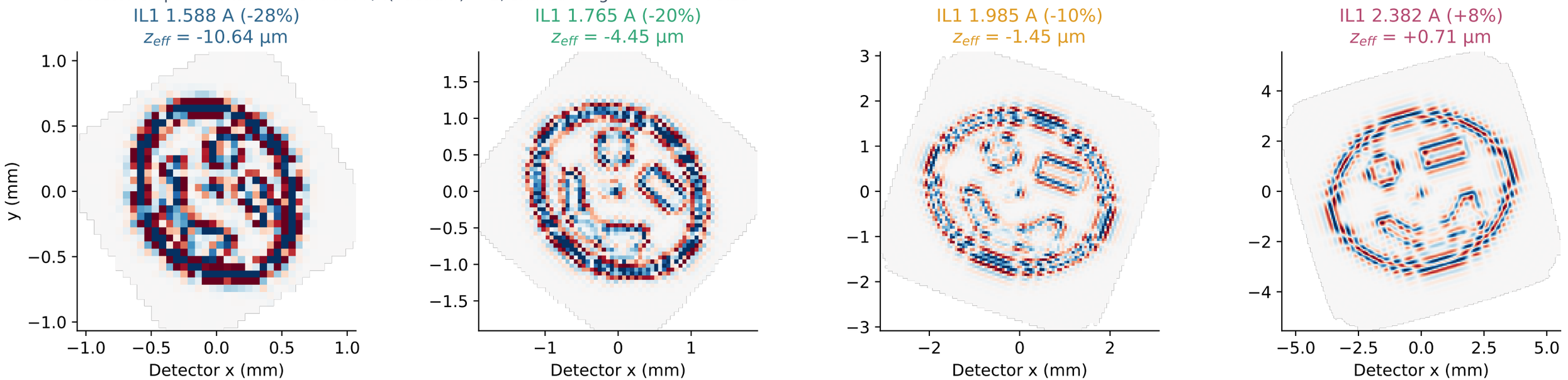
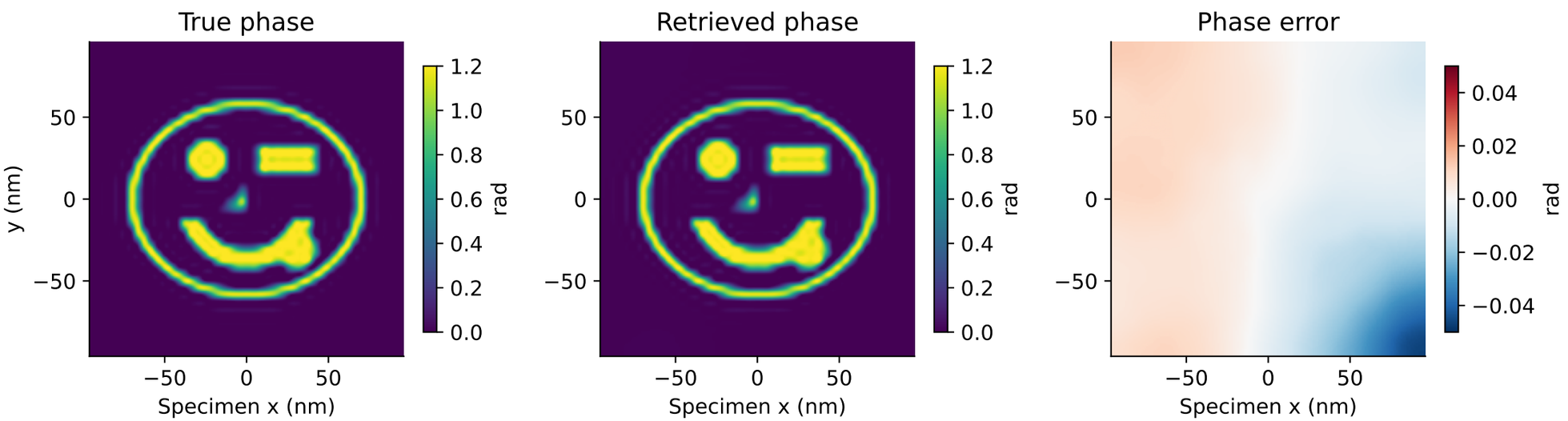
Calibrated IL1 control makes phase measurable

Four IL1 settings → calibrated Gaussian propagation → one recovered exit wave

Detector crops: normalized contrast I / I(vacuum) − 1; shared range −0.25 to +0.25





Known for each exposure

IL1 current → full optical model

Defocus + magnification + rotation

Phase RMS error
0.0068 rad

200 kV · coherent Gaussian beamlets

Ideal lenses · noiseless synthetic data

Complex wave fitted from intensity only